

### 5.3.18 Temperature sensor characteristics

**Table 55. Temperature sensor characteristics**

| Symbol                   | Parameter                                                          | Min | Typ  | Max  | Unit    |
|--------------------------|--------------------------------------------------------------------|-----|------|------|---------|
| $T_L^{(1)}$              | $V_{SENSE}$ linearity with temperature (from $V_{sensor}$ voltage) | -   | -    | 3    | °C      |
| Avg_Slope <sup>(2)</sup> | Average slope (from $V_{sense}$ voltage)                           | -   | 2.14 | -    | mV/°C   |
| $V_{30}^{(3)}$           | Voltage at 30° C ( $\pm 1$ ° C)                                    | -   | 0.65 | -    | V       |
| $t_{start\_run}^{(1)}$   | Startup time in Run mode (buffer startup)                          | -   | -    | 25.2 | $\mu$ s |
| $t_{S\_temp}^{(1)}$      | ADC sampling time when reading the temperature                     | 13  | -    | -    |         |
| $I_{sens}^{(1)}$         | Sensor consumption                                                 | -   | 0.18 | 0.29 | $\mu$ A |
| $I_{sensbuf}^{(1)}$      | Sensor buffer consumption                                          | -   | 3.8  | 6.5  |         |

1. Specified by design - Not tested in production.

2. Evaluated by characterization - Not tested in production.

3. Measured at  $V_{DDA} = 3.3$  V  $\pm 10$  mV. The  $V_{30}$  ADC conversion result is stored in the TS\_CAL1.

**Table 56. Temperature sensor calibration values**

| Symbol  | Parameter                                                               | Memory address            |
|---------|-------------------------------------------------------------------------|---------------------------|
| TS_CAL1 | Temperature sensor raw data acquired value at 30 °C, $V_{DDA} = 3.3$ V  | 0x08FF F814 - 0x08FF F815 |
| TS_CAL2 | Temperature sensor raw data acquired value at 140 °C, $V_{DDA} = 3.3$ V | 0x08FF F818 - 0x08FF F819 |

### 5.3.19 Digital-to-analog converter characteristics (DAC)

**Table 57. DAC characteristics**

Specified by design and not tested in production.

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| Symbol               | Parameter                                                | Conditions            |                               | Min  | Typ              | Max                    | Unit |
|----------------------|----------------------------------------------------------|-----------------------|-------------------------------|------|------------------|------------------------|------|
| V <sub>DDA</sub>     | Analog supply voltage                                    | -                     |                               | 2.7  | -                |                        | V    |
| V <sub>REF+</sub>    | Positive reference voltage                               | -                     |                               | 2.5  | -                | V <sub>DDA</sub>       |      |
| V <sub>REF-</sub>    | Negative reference voltage                               | -                     |                               | -    | V <sub>SSA</sub> | -                      |      |
| R <sub>L</sub>       | Resistive Load                                           | DAC output buffer ON  | connected to V <sub>SSA</sub> | 5    | -                | -                      | KΩ   |
|                      |                                                          |                       | connected to V <sub>DDA</sub> | 25   | -                | -                      |      |
| R <sub>O</sub>       | Output Impedance                                         | DAC output buffer OFF |                               | 10.3 | 13.00            | 16                     |      |
| R <sub>BON</sub>     | Output impedance sample and hold mode, output buffer ON  | DAC output buffer ON  | V <sub>DD</sub> = 2.7 V       | -    | -                | 1.6                    | KΩ   |
| R <sub>BOFF</sub>    | Output impedance sample and hold mode, output buffer OFF | DAC output buffer OFF | V <sub>DD</sub> = 2.7 V       | -    | -                | 17.8                   | KΩ   |
| C <sub>L</sub>       | Capacitive Load                                          | DAC output buffer OFF |                               | -    | -                | 50                     | pF   |
| C <sub>SH</sub>      |                                                          | Sample and Hold mode  |                               |      | 0.10             | 1                      | μF   |
| V <sub>DAC_OUT</sub> | Voltage on DAC_OUT output                                | DAC output buffer ON  |                               | 0.2  | -                | V <sub>DDA</sub> - 0.2 | V    |
|                      |                                                          | DAC output buffer OFF |                               | 0    | -                | V <sub>REF+</sub>      |      |